

face photo
 $128 \times 128 \times 3$

VGG16 convolutional base
frozen: 14,714,688 weights
output $4 \times 4 \times 512$

the same for both outputs:
each photo passes through once

Flatten: 8,192 numbers

age_hidden
Dense 256, ReLU

gender_hidden
Dense 256, ReLU

age: Dense 1, linear
loss: mean absolute error

gender: Dense 1, sigmoid
loss: binary cross-entropy